

# Primary VPT Chain: ProducerUA -> ProZorro -> Retail

## INTERPRETATION GUIDE

- ADF  $p > 0.05$  and KPSS  $p < 0.05$  -> likely I(1)-like; prefer differences/cointegration models.
- ADF  $p < 0.05$  and KPSS  $p > 0.05$  -> stationary; level models are more admissible.
- Ljung-Box  $p < 0.05$  -> autocorrelation; add lag structure.
- BP/White  $p < 0.05$  -> heteroskedasticity; use robust or HAC standard errors.
- JB  $p < 0.05$  -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

## RUN NOTES

standardized\_type=cottage\_cheese

retailer=silpo

promo\_variant=promo\_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo\_novus. Combined rule: daily median of available silpo and novus standardized\_type prices.

Table: PreTests

| series               | adf_level_p            | kpss_level_p | adf_diff1_p           | kpss_diff1_p       | adf_diff2_p           | kpss_diff2_p          | integration_class | stability_flag | standardized_type | retailer | promo_variant    | cointegration_p       | pair_n_obs |
|----------------------|------------------------|--------------|-----------------------|--------------------|-----------------------|-----------------------|-------------------|----------------|-------------------|----------|------------------|-----------------------|------------|
| producer             | nan                    | nan          | nan                   | nan                | nan                   | nan                   | ambiguous         | 0.0            | cottage_cheese    | silpo    | promo_controlled | nan                   | nan        |
| prozorro             | 1.1483227108295005e-10 | 0.01         | 4.589982408975472e-17 | 0.1                | 4.813752592550519e-10 | 0.1                   | I(1)              | 0.0            | cottage_cheese    | silpo    | promo_controlled | nan                   | nan        |
| retail               | 5.9157706543378735e-08 | 0.1          | 7.517018918246667e-13 | 0.0865089553841217 | 0.0018569176848127478 | 0.0416666666666666796 | I(0)              | 0.0            | cottage_cheese    | silpo    | promo_controlled | nan                   | nan        |
| producer_to_prozorro | nan                    | nan          | nan                   | nan                | nan                   | nan                   | pair              | nan            | cottage_cheese    | silpo    | promo_controlled | nan                   | 0.0        |
| prozorro_to_retail   | nan                    | nan          | nan                   | nan                | nan                   | nan                   | pair              | nan            | cottage_cheese    | silpo    | promo_controlled | 0.0044991066582690566 | 35.0       |
| producer_to_retail   | nan                    | nan          | nan                   | nan                | nan                   | nan                   | pair              | nan            | cottage_cheese    | silpo    | promo_controlled | nan                   | 0.0        |

Table: ModelCoefficients

| standardized_type | retailer | promo_variant    | frequency | link | model_family | y_series | x_series | n_obs | sr_coef | lr_coef | ect_coef | ect_pvalue | asymmetry_short_p | asymmetry_long_p | vecm_rank | model_status | unreliable_flag | notes                        | primary_chain_doc            |
|-------------------|----------|------------------|-----------|------|--------------|----------|----------|-------|---------|---------|----------|------------|-------------------|------------------|-----------|--------------|-----------------|------------------------------|------------------------------|
| cottage_cheese    | silpo    | promo_controlled | daily     | none | none         | n/a      | n/a      | 203   | nan     | nan     | nan      | nan        | nan               | nan              | nan       | unavailable  | 1               | No admissible model estim... | Primary chain is strictly... |

# Table: ResidualDiagnostics

| model_family | link | y_series | x_series | ljungbox_p | arch_p | jb_p | unreliable_flag |
|--------------|------|----------|----------|------------|--------|------|-----------------|
| none         | none | n/a      | n/a      | nan        | nan    | nan  | 1               |

**Table: SeriesUsed**

| date       | producer | prozorro           | retail | standardized_type | retailer | promo_variant    | combined_rule                |
|------------|----------|--------------------|--------|-------------------|----------|------------------|------------------------------|
| 2025-04-14 | nan      | 138.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-04-16 | nan      | 118.8              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-04-18 | nan      | 108.9              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-04-21 | nan      | 120.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-04-22 | nan      | 99.0               | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-04-23 | nan      | 107.08             | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-04-24 | nan      | 160.2              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-04-29 | nan      | 156.06             | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-04-30 | nan      | 226.25             | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-01 | nan      | 82.5               | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-02 | nan      | 118.08000000000001 | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-05 | nan      | 101.29499999999999 | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-06 | nan      | 115.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-07 | nan      | 258.2              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-08 | nan      | 145.4              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-09 | nan      | 101.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-11 | nan      | 89.4               | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-12 | nan      | 168.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-13 | nan      | 131.34             | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-14 | nan      | 75.0               | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-15 | nan      | 152.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-16 | nan      | 145.2              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-19 | nan      | 98.75              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-20 | nan      | 159.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-21 | nan      | 134.04             | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-22 | nan      | 139.98000000000002 | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-23 | nan      | 107.55             | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-26 | nan      | 168.6              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-27 | nan      | 156.06             | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-29 | nan      | 150.5              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-05-30 | nan      | 165.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-02 | nan      | 135.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-03 | nan      | 122.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-05 | nan      | 142.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-06 | nan      | 90.52000000000001  | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-09 | nan      | 60.0               | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-10 | nan      | 112.98             | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-11 | nan      | 93.0               | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-12 | nan      | 108.0              | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |
| 2025-06-13 | nan      | 71.5               | nan    | cottage_cheese    | silpo    | promo_controlled | daily_median_of_available... |

## Table: LagProfile

|                    | pair | lag                   | corr |
|--------------------|------|-----------------------|------|
| prozorro_to_retail | 1    | -0.03042410853195715  |      |
| prozorro_to_retail | 2    | 0.09315059121833928   |      |
| prozorro_to_retail | 3    | -0.013554687740592215 |      |
| prozorro_to_retail | 4    | 0.2640036394817208    |      |
| prozorro_to_retail | 5    | -0.01665792884614875  |      |
| prozorro_to_retail | 6    | 0.12427459978543254   |      |
| prozorro_to_retail | 7    | -0.1940209713635291   |      |
| prozorro_to_retail | 8    | 0.030775604018925283  |      |
| prozorro_to_retail | 9    | -0.09372737565447635  |      |
| prozorro_to_retail | 10   | -0.07704229808546917  |      |
| prozorro_to_retail | 11   | -0.17919361937712847  |      |
| prozorro_to_retail | 12   | 0.18757301375096241   |      |
| prozorro_to_retail | 13   | 0.10991817659598781   |      |
| prozorro_to_retail | 14   | -0.34943263469401786  |      |
| prozorro_to_retail | 15   | 0.08988836742206224   |      |
| prozorro_to_retail | 16   | 0.24534854931006433   |      |
| prozorro_to_retail | 17   | 0.08079616663444002   |      |
| prozorro_to_retail | 18   | 0.1868750222264715    |      |
| prozorro_to_retail | 19   | 0.13127219217068242   |      |
| prozorro_to_retail | 20   | 0.004357040410090344  |      |
| prozorro_to_retail | 21   | 0.08986030260644852   |      |
| prozorro_to_retail | 22   | 0.1322365935271666    |      |
| prozorro_to_retail | 23   | -0.3991469531086648   |      |
| prozorro_to_retail | 24   | 0.16752752371267343   |      |
| prozorro_to_retail | 25   | 0.05655494006971566   |      |
| prozorro_to_retail | 26   | -0.03561378213964211  |      |
| prozorro_to_retail | 27   | -0.000797420541569574 |      |
| prozorro_to_retail | 28   | 0.20625749020049242   |      |
| prozorro_to_retail | 29   | 0.06987978494683865   |      |
| prozorro_to_retail | 30   | 0.055791968171944914  |      |

Table: ModelEligibility

| standardized_type | retailer | promo_variant    | frequency | integration_producer | integration_prozorro | integration_retail | n_obs_producer | n_obs_prozorro | n_obs_retail | n_obs_pair_prod_prozorro | n_obs_pair_prozorro_retail | any_i2 | eligibility_note             |
|-------------------|----------|------------------|-----------|----------------------|----------------------|--------------------|----------------|----------------|--------------|--------------------------|----------------------------|--------|------------------------------|
| cottage_cheese    | silpo    | promo_controlled | daily     | ambiguous            | I(1)                 | I(0)               | 0              | 190            | 48           | 0                        | 35                         | 0      | Low pair overlap in main ... |

**Table: NARDL\_Multipliers**

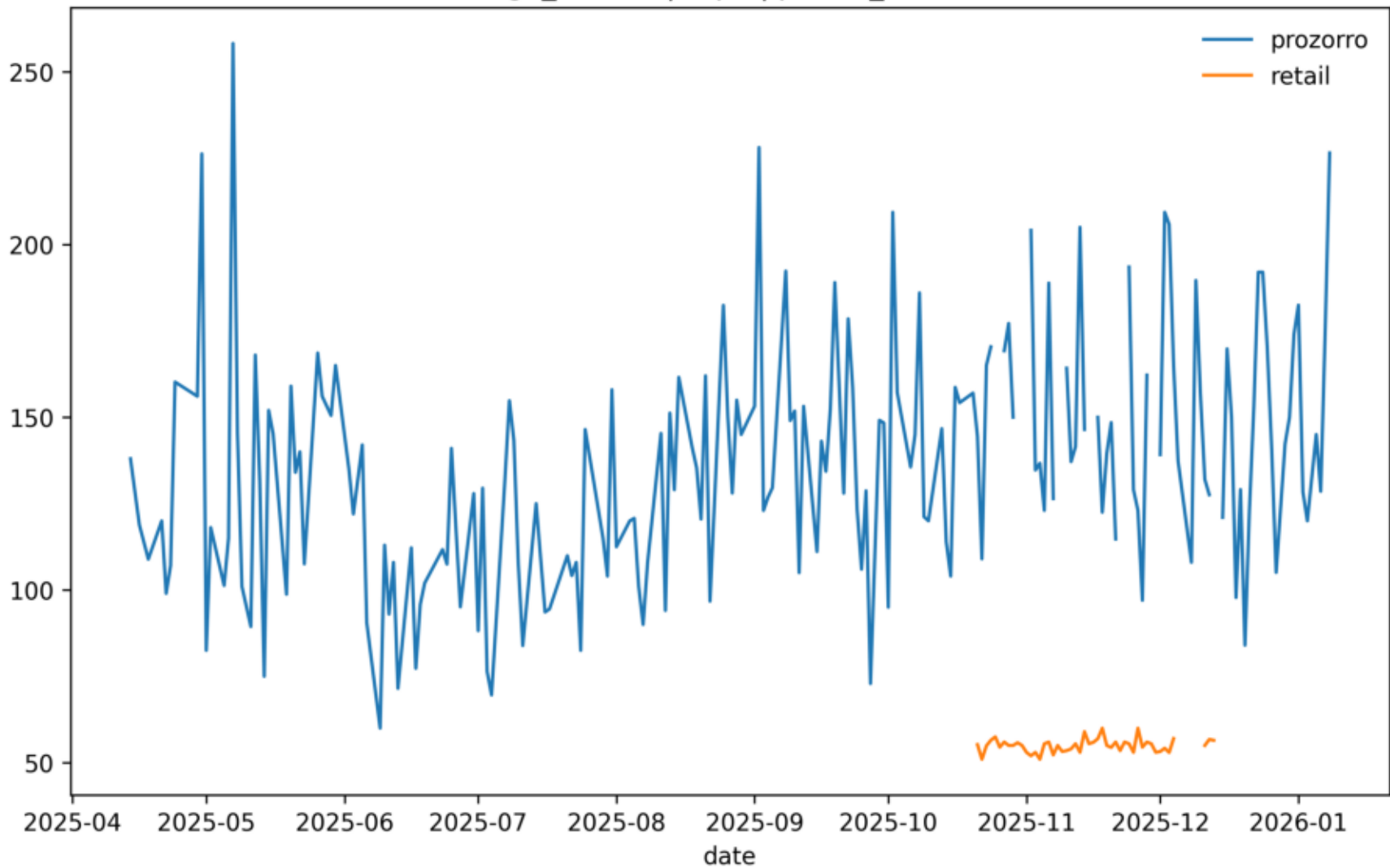
No rows



**Table: VECM\_IRF**

No rows

time\_series\_promo\_controlled.png  
cottage\_cheese | silpo | promo\_controlled



## Lag-Correlation Profile | cottage\_cheese | silpo | promo\_controlled

